

coverage.vr

File location: \$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/coverage/circ_bfr/coverage.vr

```
#include <vera_defines.vrh>

class cb_coverage {
    local bit [3:0] rd_ptr;
    local bit [3:0] wr_ptr;
    event cover_it;

    coverage_group buffer_occupancy {
        sample_event = sync(ALL, this.cover_it) async;
        sample level (('h10 + wr_ptr - rd_ptr) % 'h10) {
            cov_comment = "Buffer occupancy level";
            state empty    (0);
            state full     ('hF);
            state occupied (not state);
            cov_weight = 0;
        }
        sample rd_offset (rd_ptr) {
            cov_comment = "Position of the RD pointer in buffer";
            state at_min (0);
            state at_max (4'hF);
            state middle (not state);
            cov_weight = 0;
        }
        sample relation (rd_ptr) {
            cov_comment = "Relative position of the RD pointer wrt WR pointer";
            state ahead  (not state) if (wr_ptr < rd_ptr);
            state behind (not state) if (wr_ptr > rd_ptr);
        }
        cross goal (level, rd_offset);
        cov_weight = 2;
    }
}

task cover(bit [3:0] rd_ptr,
          bit [3:0] wr_ptr) {
    this.rd_ptr = rd_ptr;
    this.wr_ptr = wr_ptr;
    trigger(this.cover_it);
}

}
```